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DECLARATION

I, Adnaan Osman, declare that this project is my original work and has not been copied from any other source. This document has been compiled and researched by me to the best of my ability.

Signature: _____

Date: _____

DEDICATION

This project is dedicated to my family for their unwavering support and inspiration. Their encouragement and belief in my abilities have been instrumental in the realization of this project.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my supervisors, family and friends for their constant encouragement and belief in my vision. Above all, I thank God for providing me with the strength and inspiration to pursue this endeavour.

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Abstract

Farmnet Kenya is an upcoming web-based platform designed to revolutionize the agricultural sector by enhancing communication and providing timely information to farmers across the country. Recognizing the challenges posed by traditional methods—such as delays and inaccuracies in information dissemination—Farmnet Kenya aims to offer real-time weather updates, facilitate seamless communication with agronomists, and provide access to valuable agricultural resources. By integrating secure payment systems like M-Pesa, the platform will streamline transactions for premium services. The primary objective is to empower farmers with the tools and knowledge necessary to make informed decisions, thereby improving productivity and contributing to the overall growth of Kenya's agricultural sector.

Farmers will be able to get services like soil testing, pest and disease management and many more.

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CHAPTER ONE: INTRODUCTION

1.1 Introduction

Farmnet Kenya is a web-based platform designed to address the communication and information challenges faced by farmers in Kenya . The platform aims to provide real-time weather updates and facilitate communication with agronomists, addressing the inefficiencies seen in traditional communication methods, such as word-of-mouth (WeatherBug, n.d.). Currently, many farmers rely on inefficient methods, such as word-of-mouth and traditional media, which often result in delays and inaccuracies. Farmnet Kenya aims to streamline these processes by offering a centralized and accessible system.

1.2 Background of the Study

1.2.1 Background of the Organization

Farmnet Kenya was conceptualized in response to the growing need for accessible agricultural information and expert advice. It was founded to bridge the gap between farmers and agronomists, ensuring better decision-making and productivity. The organization operates from Nairobi, Kenya, and focuses on improving the livelihoods of farmers through innovative technology solutions. Over the years, its growth has highlighted the need for a robust system to manage communication and information dissemination.

1.2.2 Overview of Existing System

The current system relies on manual methods, such as physical meetings and phone calls, to disseminate information. Farmers often face delays in receiving critical weather updates, and access to agronomists is limited due to logistical challenges. These inefficiencies lead to poor crop planning and reduced agricultural productivity.

1.2.3 Overview of Proposed System

The proposed system, Farmnet Kenya, will provide a web-based platform where farmers will:

Receive real-time weather updates.

Communicate securely with agronomists.

Access agricultural resources and expert advice.

Utilize M-Pesa for secure payments.

Access agronomist services to improve farming practices.

This system will enhance efficiency, reduce delays, and improve overall productivity.

1.3 Problem Statement

The agricultural sector in Kenya faces several challenges due to the lack of timely and reliable information. Farmers often struggle with:

Delayed or inaccurate weather updates, leading to poor planning.

Limited access to agronomists for expert advice.

Inefficient communication methods that waste time and resources.

These problems hinder productivity and reduce the potential for growth in the sector.

Farmnet Kenya intends to address these issues by providing a centralized solution for real-time information and seamless communication.

1.4 Objectives

1.4.1 General Objective

To develop a web-based platform that empowers farmers with tools to improve decision-making, enhance communication, and facilitate secure payments for agronomist services, thereby providing timely agricultural information and support.

1.4.2 Specific Objectives

To implement a farmer and agronomist registration system for personalized recommendations and services tailored to individual needs.

To develop a real-time weather update module that will enable farmers to plan their activities effectively based on accurate and timely weather information.

To design a secure messaging tool that will enhance communication between farmers

and agronomists throughout the year, fostering better collaboration.

To integrate a payment system that allows seamless transactions between farmers, agronomists, and other stakeholders, supporting financial transparency.

To create a reporting tool that provides comprehensive summaries of farming activities to both farmers and agronomists for informed decision-making.

1.5 Justification

Farmnet Kenya will be a timely solution to the challenges facing the agricultural sector.

By providing a centralized platform, it will:

Empower farmers with reliable and timely information.

Enhance decision-making through direct access to expert advice.

Improve agricultural productivity, benefiting the entire value chain.

The platform's innovative approach ensures its relevance and value in the current technological landscape.

1.6 Scope of the Study

The project will focus on developing and implementing key features, including:

Real-time weather updates module.

Messaging capabilities for farmer-agronomist communication module.

Integration with M-Pesa for secure transactions module.

Summary reporting of farming activities module.

Farmer and agronomist registration for personalized services module.

Out of Scope:

Advanced analytics for predictive farming.

Supply chain management features.

This ensures the project goals are practical and achievable within one academic year.

1.7 Limitations of the Proposed System

Limited financial resources may restrict the implementation of advanced features.

Time constraints due to academic deadlines.

Lack of extensive programming knowledge, which may necessitate external support.

1.8 Project Risk and Mitigation

Table 1.1 Risks and mitigations

1.9 Project Schedule

The project will be executed in phases, including research, design, development, testing, and deployment

Research: 2 months (Dec-Jan)

Design: 3 months (Feb-Apr)

Development: 4 months (Apr-Jul)

Testing: 2 months (Jul-Aug)

Deployment: 1 month (Aug)

Figure 1.1 Project Schedule Gantt chart

1.10 Budget and Resources

To ensure the successful development and implementation of Farmnet Kenya, the following resources and budget considerations are necessary:

Table 1.2 Budget and Resources

Chapter Two: Literature Review

2.0 Introduction

This chapter consists of an exploration of what has been done in the area of study, including related works to this study and an analysis of why their results are not being adopted or accepted for the current project. The focus is on peer-reviewed papers and validated solutions, ensuring credibility and relevance.

Farmnet Kenya aims to support smallholder farmers by providing access to weather updates and connecting them with agronomists to improve farm productivity and market access. Much like the findings in a study on agricultural decision support tools (DSTs), where advisors play a key role in promoting the use of DSTs, Farmnet Kenya could benefit from leveraging advisors as "change agents" to encourage adoption of digital tools among farmers, ensuring they can adapt to the challenges of climate change and improve their farm operations .

Farmnet Kenya can provide tailored weather forecasts and agricultural advisories to farmers, allowing them to make informed decisions about planting, harvesting, and irrigation. This aligns with the article's focus on using agrometeorological data for improved farming outcomes. The article discusses how platforms can support farmers by delivering actionable climate and weather data to improve agricultural productivity .

2.1 Reviewed Similar Systems

2.1.1 Agworld

Agworld is a farm management software that helps farmers track and manage their operations. It offers tools for farm management, weather forecasting, financial tracking, and collaboration with agricultural advisors. Although Agworld provides valuable functionalities for farmers, it is not focused on the Kenyan market, and its availability in Kenya is limited

Strengths

Agworld offers a comprehensive suite of features, including weather forecasting, farm management, and collaboration tools.

The platform helps farmers make data-driven decisions, optimizing resources and improving yields.

Weaknesses

Limited Focus on Kenya: Agworld has not been widely adopted in Kenya and is primarily available in regions like Australia, the US, and Canada. Its services are not specifically tailored to the Kenyan market.

Lacks Payment Integration: The platform does not integrate with mobile payment solutions like M-Pesa, which are crucial for farmers in Kenya.

2.1.2 Granular

Granular is a farm management software designed to help farmers optimize their operations through data-driven decisions. It provides tools for tracking crop performance, managing input costs, and analyzing farm financials. Granular aims to improve profitability by streamlining farm operations and offering insights into efficiency .

Strengths

Data-Driven Insights: Granular provides powerful analytics that helps farmers optimize their operations and improve profitability.

Resource Efficiency: By tracking inputs like machinery and labor, Granular helps farmers reduce waste and increase productivity.

Weaknesses

Limited Availability in Kenya: Similar to Agworld, Granular has not been widely adopted in Kenya, and its features may not be tailored to the specific needs of Kenyan farmers.

High Cost: Granular's pricing may be prohibitive for smallholder farmers in developing countries like Kenya.

2.1.3 WeatherBug

WeatherBug is a weather tracking and forecasting platform that provides real-time updates, severe weather alerts, and hyperlocal forecasts. While not specifically designed for farmers, it offers valuable weather data that can inform agricultural

planning (WeatherBug, 2021).

Strengths

Hyperlocal Forecasting: WeatherBug provides accurate, localized weather forecasts that can help farmers plan their activities effectively.

Real-Time Alerts: Farmers can receive timely alerts about severe weather conditions, enabling them to take preventive measures.

Weaknesses

Not Agriculture-Specific: WeatherBug is a general-purpose weather platform and lacks features tailored to the specific needs of farmers.

No Integration with Farm Management: The platform does not integrate with farm management tools or provide advice on how weather conditions impact agricultural activities.

2.1.4 Farmers Business Network (FBN)

FBN is a network-driven platform that connects farmers to share data, access inputs, and improve farm performance. It provides insights on crop performance, competitive pricing for inputs, and peer benchmarking .

Strengths

Data Sharing and Benchmarking: FBN allows farmers to compare their performance with peers, helping them identify areas for improvement.

Cost Efficiency: Farmers can access competitively priced inputs through the network, reducing production costs.

Community Support: The platform fosters collaboration among farmers, enabling them to share knowledge and experiences.

Weaknesses

Limited Focus on Africa: FBN is largely targeted at farmers in the US and Canada, with minimal adoption or support in African markets.

Requires Internet Connectivity: The platform relies heavily on internet access, which may be a limitation for farmers in remote areas of Kenya.

2.1.5 WhatsApp

WhatsApp is a widely used messaging application that allows users to send text messages, voice messages, and multimedia, as well as make voice and video calls. Its simplicity and widespread adoption make it an effective tool for real-time communication.

Strengths:

User-Friendly: The platform is easy to use and accessible on both smartphones and desktops.

Wide Adoption: With a large user base, it's familiar to many and does not require additional training.

Secure Communication: Offers end-to-end encryption, ensuring private and secure messaging.

Weaknesses:

Not Agriculture-Specific: While useful for communication, it lacks specific agricultural tools and integrations.

2.1.6 Salesforce

Salesforce is a customer relationship management (CRM) platform that provides tools for data management, analytics, and collaboration. It can be adapted for agricultural advisory services by leveraging its robust data-sharing capabilities.

Strengths

Customizable: Salesforce can be tailored to meet the unique needs of different industries, including agriculture.

Data Insights: The platform offers advanced analytics to help organizations make data-driven decisions.

Weaknesses:

High Cost: Salesforce's pricing may be prohibitive for small-scale users.

Complex Setup: The platform requires significant customization and expertise to implement.

2.2 Tools and Methodologies Used in Reviewed Systems

2.2.1 Agworld

1. **Mobile App:** Access to real-time data and farm activity tracking.
2. **Cloud-Based System:** Facilitates data storage and sharing across devices.
3. **Weather API Integration:** Provides localized weather data.

2.2.2 Granular

1. **Farm Analytics Dashboard:** Aggregates and analyzes data on crop yields and resource usage.
2. **Resource Management Tools:** Manages labor, machinery, and other farm inputs.
3. **Financial Management Tools:** Tracks costs and offers profitability analysis.

2.2.3 WeatherBug

1. **Weather Sensors and APIs:** Provides real-time weather data and hyperlocal forecasts.
2. **Alert Systems:** Sends notifications for severe weather conditions.

2.2.4 FBN

1. **Data Analytics:** Provides insights on crop performance and benchmarking.
2. **Marketplace for Inputs:** Offers a platform for farmers to buy competitively priced inputs.
3. **Peer Networking:** Connects farmers to share knowledge and experiences.

2.2.5 WhatsApp

1. **WhatsApp Business API:** Enables automated customer service, sending notifications, and scaling communication with customers.
2. **Chatbots and Automation Tools:** Integrates AI-powered chatbots to automate

responses, lead generation, and customer service.

3. WhatsApp Marketing Tools: Includes broadcast lists and CRM integrations for targeted communication and marketing campaigns.

2.2.6 Salesforce

1. Salesforce Sales Cloud: A comprehensive CRM tool for managing sales operations, tracking interactions, and analyzing customer data.

2. Salesforce Service Cloud: Manages customer service operations, integrating communication across multiple channels like WhatsApp.

3. Salesforce Marketing Cloud: Automates and tracks marketing campaigns across various channels, including email, social media, and customer engagement tools.

2.3 Gaps in the Existing Systems and the Proposed Solution

2.3.1 Gaps Identified

1. Limited Payment Integration: None of the reviewed systems, including Agworld and Granular, integrate mobile payment systems like M-Pesa, which are critical for farmers in Kenya.

2. Lack of Focus on Kenya: The reviewed systems are primarily designed for developed markets and fail to address the unique needs of Kenyan farmers, such as localized advice and affordability.

3. Disjointed Features: Platforms like WeatherBug and FBN offer specialized features but lack comprehensive farm management solutions that integrate weather data, financial management, and expert collaboration in one platform.

2.3.2 Farmnet Kenya's Solution

1. M-Pesa Integration: To address the lack of payment solutions, Farmnet Kenya will integrate M-Pesa for seamless financial transactions.

2. Localized Features: The platform will be tailored to the needs of Kenyan farmers, including localized weather forecasts and advice from agronomists.

3. Comprehensive Farm Management: Unlike disjointed systems, Farmnet Kenya will combine weather tracking, financial management, and expert collaboration into a single platform.

2.4 The Proposed Solution

Farmnet Kenya will bridge the gaps in existing systems by:

1. Proposed Security Features: Incorporating two-factor authentication, encrypted communications, and secure payment gateways to protect user data and transactions.
2. Technologies for Implementation: Leveraging PHP for backend development, JavaScript for frontend interactivity, MySQL for database management, and APIs for weather data integration.
3. Development Approach: Adopting an agile methodology to ensure continuous user feedback and iterative improvements during the development process.
4. Web and Mobile Integration: Employing responsive design principles to enable seamless synchronization and functionality across web and mobile platforms.

CHAPTER THREE: METHODOLOGY

3.1 Methodology and Tools

The system development methodology adopted for Farmnet Kenya is the Agile Development Model. This model supports iterative, user-driven development through short, manageable cycles known as sprints. It is ideal for this project because it allows feedback from farmers and agronomists to be collected regularly and used to make timely improvements to the platform.

Reasons for Choosing Agile Methodology

Enables continuous engagement with users through interviews and testing.

Supports flexible and iterative development — critical when time and resources are limited.

Encourages early delivery of core features like weather updates and messaging.

Suitable for a minimum viable product (MVP) approach.

3.1.1 Phases of the Agile Methodology:

1. Requirement Gathering:

The system will conduct user stories (short interviews/surveys) with farmers and agronomists to define features (e.g., "As a farmer, I require real-time weather updates to plan for harvest").

It will also prioritize features in a product backlog (e.g., M-Pesa integration > weather API > messaging).

This will be implemented by

Surveys: Use Google Forms (shared via WhatsApp/SMS for accessibility) to gather farmer needs in 2-week sprints.

Interviews: Hold weekly calls with agronomists to refine requirements iteratively.

2. Design:

The system will create low-fidelity wireframes (Figma) in sprints, validated by users before coding.

Also, it will use feedback loops: Show prototypes to farmers per sprint for quick iterations.

For example,

Design the "weather module" UI in Sprint 1, test with farmers, and refine in Sprint 2 based on their input (e.g., add Swahili language support).

3. Development:

It will develop minimum viable features (MVFs) per sprint (e.g., Sprint 1: User registration; Sprint 2: Basic messaging).

It will also use daily standups to track blockers (e.g., "M-Pesa API integration delayed—need alternative payment testing").

For example, Tech Stack:

Sprint 1: PHP/MySQL backend for farmer profiles.

Sprint 2: JavaScript frontend for weather alerts.

4. Testing:

Testing will be done per sprint instead of at the end. Farmers validate functionality early.

An example is

Sprint 3: Test weather API with farmers—fix inaccurate location data in the next sprint.

Use Postman for API tests; BrowserStack for mobile compatibility checks.

5. Deployment:

Features will be deployed as they are completed (e.g., release MVP with weather + messaging first).

GitHub Pages will be used for staging; gather feedback before final launch.

6. Maintenance:

After launch, we will use sprint retrospectives to prioritize fixes (e.g., "Add Swahili chatbot in next sprint").

Monthly surveys will be used to identify new needs.

3.1.2 Tools to Be Used:

1. Design Tools:

1. Requirement Gathering (Weeks 1-2)

Trello for backlog prioritization.

User stories to document needs (e.g., "As a maize farmer, I need SMS weather alerts to plan irrigation").

2. System Design (Weeks 3-4)

Figma for UI/UX prototypes.

Lucidchart for:

Entity-Relationship Diagrams (ERDs): Database structure.

Data Flow Diagrams (DFDs): System processes.

3. Development (Weeks 5-10)

Frontend: Bootstrap/jQuery (responsive design).

Backend: PHP/MySQL (cost-effective for Kenyan infrastructure).

APIs: OpenWeatherMap (weather), Safaricom M-Pesa (payments).

4. Testing (Weeks 11-12)

Techniques:

Unit Testing: Postman for API validation.

3.2 Source of Data

The data for Farmnet Kenya will be collected from both primary and secondary sources:

3.2.1 Primary Data:

1. Farmers: Information on their needs, challenges, and preferences.
2. Agronomists: Insights on the type of advice and services farmers require.
3. Agricultural Organizations: Data on existing systems and gaps in the market.

3.2.2 Secondary Data:

1. Weather Data: From free APIs like OpenWeatherMap.
2. Agricultural Reports: From organizations like the Food and Agriculture Organization (FAO) and the Kenya Agricultural and Livestock Research Organization (KALRO).
3. Existing Systems: Analysis of platforms like Agworld, Granular, and FBN to identify gaps.

3.3 Data Collection Methods

To gather the necessary data, the following methods will be used:

3.3.1 Interviews:

Conduct one-on-one interviews with farmers and agronomists to understand their needs and challenges.

Use open-ended questions to gather detailed insights.

3.3.2 Questionnaires:

Distribute online surveys (using Google Forms) to a larger group of farmers and agronomists.

Focus on quantitative data, such as the frequency of weather updates needed and preferred communication methods.

3.3.3 Observation:

Observe how farmers currently access weather information and communicate with agronomists.

Identify pain points in their current processes.

3.3.4 Document Inspection:

Review existing agricultural reports, weather data, and case studies to gather secondary data.

3.3.5 Participation:

Engage with agricultural communities and forums to gather insights and feedback.

3.4 Resources Required / Materials

3.4.1 Hardware Specifications:

Table 3.1Hardware Specifications

3.4.2 Software Specifications:

Table 3.2 Software Specifications

3.5 Time Schedule

The project will be executed in the following phases:

Table 3.3 Project Schedule

3.6 Project Cost

Table 3.4 Project Cost

3.7 Expected Final Product

The final product will be a web-based platform with the following features:

Real-time weather updates.

Secure messaging between farmers and agronomists.

M-Pesa integration for payments.

Reporting tools for farming activities.

Contribution:

The platform will provide a practical solution to the challenges faced by farmers in Kenya.

It will serve as a model for similar initiatives in other developing countries.

The results of the project will be shared through a conference paper or publication in an agricultural or technology journal.

CHAPTER FOUR: SYSTEM ANALYSIS AND REQUIREMENT MODELING

4.1 Introduction to the system analysis

This chapter presents a detailed analysis of the existing agricultural information and

communication practices used by farmers in Kenya, as well as the system requirements and models for the proposed Farmnet Kenya platform. System analysis involves understanding and documenting the current problems in the agricultural support structure and how technology can be used to improve information delivery, expert support, and payment systems.

The analysis process includes gathering facts from both primary and secondary data sources, identifying functional and non-functional requirements, and modeling the proposed solution using various system analysis tools such as Data Flow Diagrams (DFDs), context diagrams, use case diagrams, and flowcharts.

Additionally, this chapter explores the feasibility of implementing the proposed system through technical, economic, operational, and scheduling feasibility studies. The goal is to determine whether the proposed platform is practical, achievable, and beneficial to its intended users—primarily smallholder farmers and agronomists. This chapter sets the foundation for the design and development of Farmnet Kenya by ensuring that the system's goals are aligned with real-world needs and supported by technical capability.

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